Merjem Memic

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Education

University of Michigan — **Ann Arbor, MI**
B.S. in Data Science — *Expected May 2025*

- **GPA:** 3.80/4.00 | **Awards:** Dean's List, Academic Scholarship
- **Courses:** Data Structures & Algorithms, Practical Data Science, Web Systems, Machine Learning, Data Mining, Linear Regression, Statistics and Probability
- **Activities:** Muslim Engineering Society (Professional Development Chair), Michigan Data Science Team (Member)

Skills

Languages: Python, C++, R, HTML/CSS/JavaScript, SQL
Technologies: PyTorch, Pandas, Beautiful Soup, Tableau, ReactJS, NodeJS, OpenAI API, Material UI, Microsoft Office
Interests: Hiking through the mountains and enjoying the open air, and crocheting small animals

Experience

University of Michigan — **Ann Arbor, Michigan**
EECS 183 Grader — *Sept 2024 – Present*

- Reviewed, graded, and provided feedback to a class of 900+ students in their first computer science class to promote good programming habits

University of Michigan — **Ann Arbor, Michigan**
Computer Science Engineering Mentor — *Sept 2024 – Present*

- Supported 20+ undergraduate computer science students in their new class and the computer Science community at Umich

Headstarter AI — **Ann Arbor, Michigan**
Fellow — *Jul 2024 – Sept 2024*

- Built a basic Pantry Manager Application using Firebase, Next.js, Material UI, and React to seamlessly allow control of pantry items
- Programmed a Library Chatbot using the Gemini API on the backend and Next.js for the frontend

EcoData — **Ann Arbor Michigan**
Frontend Developer — *Jan 2024 – Apr 2024*

- Built a user-friendly and interactive page for a free website to help individuals plan meals without excessive spending and food waste

Projects

Assessing Harmful Algal Blooms in Lake Erie

- Modeled multiple data sets of algae producing nutrients collected across Lake Erie using data visualization tools to realize an increase in the severity of blooms after COVID-19
- Utilized Python libraries Numpy, matplotlib, and scikit-learn to perform statistical analysis discovered a high correlation between nutrient content and water runoff
- Produced scatter plots, line plots, and heat maps with Python functions to visualize results to a non-technical audience and describe the future problems facing Lake Erie
- Cleaned data to ensure accuracy of observations and seamless integration into analysis

Comparative Analysis of CNN, RNN, and SVM for Smartphone-Based Human Activity Recognition

- Implemented a Support Vector Machine, Convolution Neural Network, and Recurrent Neural Network to classify a person's activities based on sensor data collected from a smartphone's embedded accelerometer and gyroscope.

Factors in the Severity of Vehicle Crashes

- Modeled large data set of information on vehicle crashes which resulted in finding a daily trend in crash severity
- Utilized R functions to calculate the average severity of crashes for all seasons to detect a pattern of more severe crashes in the winter months on while more crashes overall happened in dry months

Euchre Interactive Card Game Interface

- Designed a C++ interface that allows users to play a game of euchre with simulated players executing moves via a programmed strategy through object-oriented programming